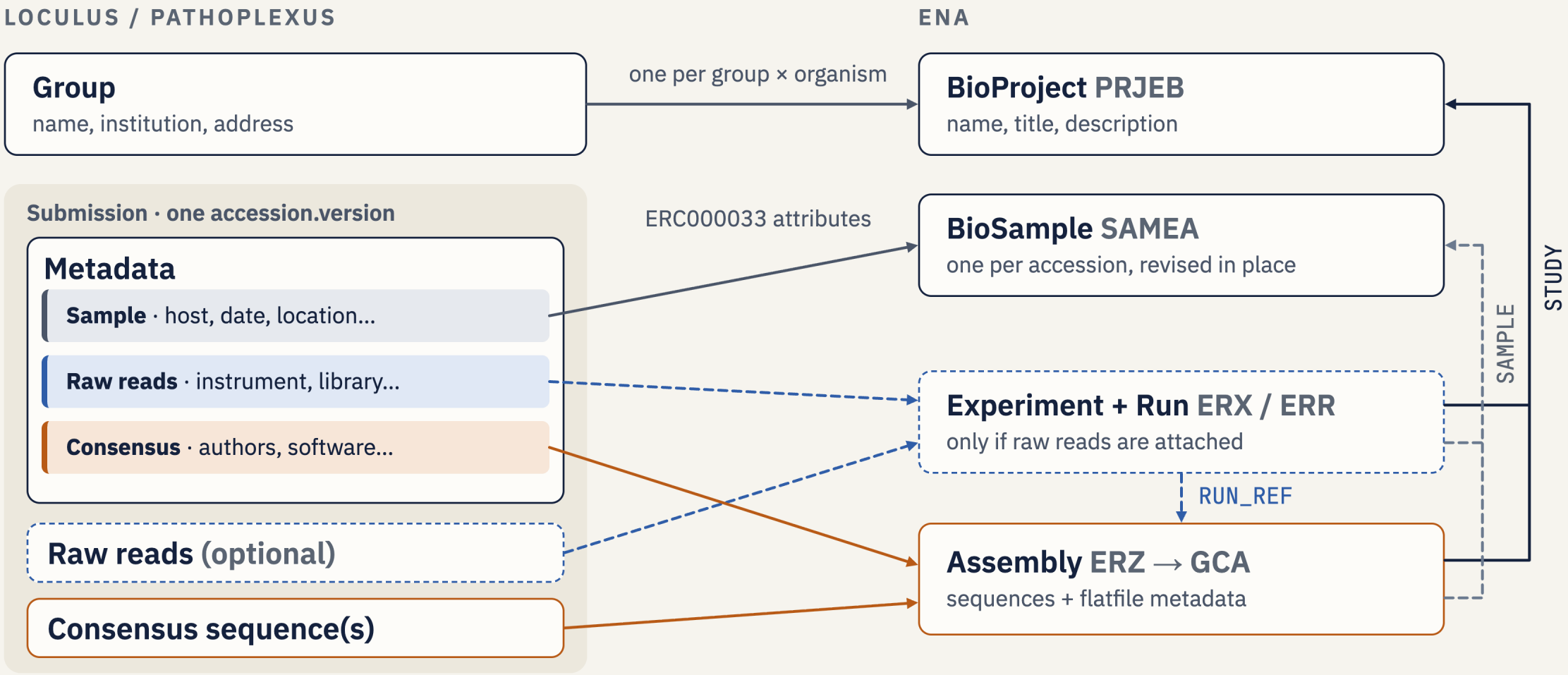
TRAINING

The ENA deposition pipeline

From a Loculus / Pathoplexus submission to BioProject, BioSample, Run and Assembly in ENA

How Loculus data maps onto ENA



What we send to ENA

HTTP POST · XML

BioProject & BioSample

Multipart POST to the Webin drop-box/submit endpoint with basic auth

submission.xml
ADD (or MODIFY) + HOLD date

project.xml / sample.xml
PROJECT_SET or SAMPLE_SET

```
<SAMPLE_ATTRIBUTE>
<TAG>collection date</TAG>
<VALUE>2023-08-26</VALUE>
```

→ **receipt XML: PRJEB… / SAMEA…**

Raw reads

`webin-cli -context reads`

`manifest.tsv` `*.fastq.gz (1, or 2 if paired)`

```
STUDY SAMPLE NAME INSTRUMENT|PLATFORM LIBRARY_SOURCE
LIBRARY_SELECTION LIBRARY_STRATEGY INSERT_SIZE FASTQ
```

→ **ERX experiment + ERR run**

Assembly

`webin-cli -context genome`

`manifest.tsv` `sequences.embl.gz` `chromosome_list.gz`

```
STUDY SAMPLE ASSEMBLYNAME COVERAGE PROGRAM PLATFORM RUN_REF
AUTHORS ADDRESS FLATFILE CHROMOSOME_LIST
```

→ **ERZ now, after processing (48h+): GCA + per-segment accessions**

What happens on a revision

A new version is its own (accession, version) row and runs all four steps again, without waiting for the previous version to finish.

BIOPROJECT

Reused

One per group × organism, never revised. The revision gets the PRJEB accession immediately.

BIOSAMPLE

MODIFY

Same alias → same SAMEA accession (biosample is not versioned on ENA). Modify is fine even if nothing changed.

RUN

New only if raw read files changed

Same files: copy last results. New files: new ERR, plus a Slack alert listing old runs to suppress.

ASSEMBLY

New only if flatfile changes

Unchanged: copy last results. Changed flatfile: new ASSEMBLYNAME and ERZ. Changed manifest: needs manual intervention - email ENA (A new RUN_REF counts as a change.)

⚠ **HAS_ERRORS, needs a manual revision:** linked biosample/bioproject changed between versions · raw reads metadata changed without new files · assembly manifest changed

The pipeline at a glance

